

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	T Jyothi Reddy	Reg. No.:	192472079
Section / Batch:	2024	Date:	

Code of Conduct

I _____ T Jyothi Reddy(192472079)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ T Jyothi Reddy _____

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations
- Conditions

- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation
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Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	3	3	4	3	3	4	20	94
2	3	3	4	4	3	3	20	
3	4	3	3	3	3	3	19	
4	3	3	4	3	3	3	19	
5	3	3	3	4	3	3	19	

Faculty In-charge	Course Coordinator	Program Director
Dr. T. Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. AI-Powered Insurance Claim Processing System

a) Structural Ambiguity Using CFG

Sentence:

“I reported the damage to the car after the accident.”

The ambiguity occurs because the prepositional phrase (PP) “**to the car**” can be attached to either the verb “**reported**” or the noun “**damage**.”

A simplified CFG is:

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$NP \rightarrow Det N \mid Det N PP$

$PP \rightarrow P NP$

Interpretation 1:

$S \rightarrow NP [VP \text{ reported } [NP \text{ the damage } [PP \text{ to the car }] [PP \text{ after the accident }]]$

Meaning: The claimant reported the damage, and the report was made to the car.

Interpretation 2:

$S \rightarrow NP [VP \text{ reported } [NP \text{ the damage } [PP \text{ to the car }]] [PP \text{ after the accident }]]$

Meaning: The damage was specifically damage to the car, and it was reported after the accident.

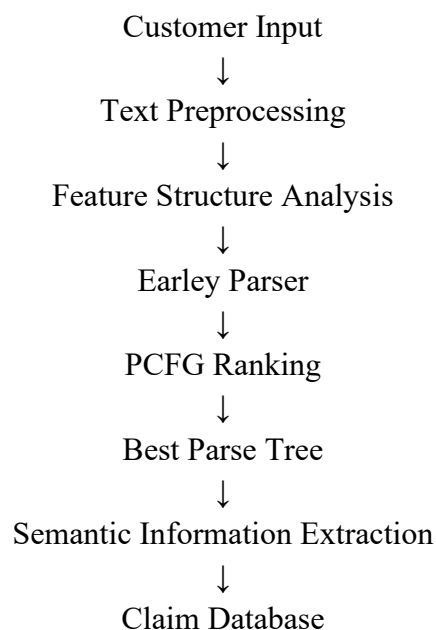
Therefore, the ambiguity is mainly caused by **PP attachment**.

b) Limitations of Basic CFG

1. Basic CFG can generate multiple valid parse trees but cannot easily decide which interpretation is correct.
2. CFG does not naturally enforce subject–verb agreement such as “customer reports” versus “customers report.”
3. Long customer descriptions can create many possible parse trees, increasing computational cost.
4. Conversational and incomplete statements may not match the predefined grammar rules.
5. CFG does not assign probabilities to competing interpretations.

c) Improved Parsing Approach

The proposed architecture is:



Feature Structures handle agreement constraints such as number and person.

Earley Parsing efficiently handles different possible structures and incomplete input.

PCFG assigns probabilities to possible parse trees and selects the most likely interpretation.

For example:

$P(\text{Parse 1}) = 0.30$

$P(\text{Parse 2}) = 0.70$

The system selects Parse 2 because it has the higher probability.

d) Justification

The proposed architecture improves:

- **Accuracy:** PCFG ranks competing interpretations and Feature Structures enforce grammatical agreement.
- **Efficiency:** Earley Parsing handles long and incomplete sentences more effectively.
- **Scalability:** The parser can process different customer-generated descriptions.
- **Reliability:** Semantic extraction is performed only after selecting the most suitable parse.
- **Automation:** Important claim details such as claimant, damaged object, cause, and amount can be extracted more consistently.

Thus, combining CFG, PCFG, Feature Structures, and Earley Parsing provides a more accurate and reliable insurance claim processing system.

Q2. Intelligent Railway Ticket Booking Assistant

a) Possible Syntactic Interpretations Using CFG

Sentence:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

A simplified CFG is:

$S \rightarrow VP$

$VP \rightarrow V \text{ NP PP PP}$

$NP \rightarrow \text{Num } N$

$PP \rightarrow P \text{ NP}$

$NP \rightarrow \text{Det } N \mid \text{Adj } N$

Possible interpretation 1:

Book [two tickets] [from Chennai to Delhi] [for my parents] [with AC sleeper].

Here, “**with AC sleeper**” describes the ticket/class information.

Possible interpretation 2:

Book [two tickets] [from Chennai to Delhi] [for my parents with AC sleeper].

Here, the parser may incorrectly attach “**with AC sleeper**” to “my parents.”

The correct interpretation is that **AC sleeper is the ticket/class attribute**, not a property of the parents.

b) Limitations of Top-Down Parsing

Top-down parsing starts from the start symbol and predicts possible grammar rules.

Problems include:

1. **Excessive backtracking:** The parser may try several rules before finding the correct interpretation.
2. **Ambiguous attachment:** Phrases such as “with AC sleeper” can have multiple possible attachments.
3. **Incomplete requests:** Spoken queries may stop before a complete grammatical structure is formed.
4. **Long-distance dependencies:** Important words may be separated by several phrases.
5. **Slow processing:** Repeated prediction and backtracking increase response time.

Therefore, basic top-down parsing is not ideal for real-time conversational ticket booking.

c) Earley Parsing

Earley Parsing is suitable because it maintains multiple possible partial parse states.

For the given sentence:

Input → Book two tickets from Chennai to Delhi for my parents with AC sleeper

Earley Parser:

1. Starts with the beginning of the sentence.
2. Predicts possible grammar structures.
3. Scans each input word.
4. Maintains incomplete structures.
5. Completes valid structures.
6. Keeps competing interpretations when ambiguity exists.
7. Selects the best interpretation using additional constraints or probabilities.

It can therefore handle:

- Ambiguous structures
- Incomplete input
- Partial sentences
- Multiple parse trees
- Long-distance dependencies

d) Comparison

Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Ambiguity Handling	Limited	Produces alternatives	Handles alternatives effectively
Backtracking	High	Depends on parser	Reduced through chart parsing
Partial Input	Poor	Limited	Good
Complex Queries	Slow	Can become expensive	More suitable
Real-Time Use	Less suitable	Moderate	Highly suitable
Incomplete Sentences	Poor	Limited	Good

Conclusion

Earley Parsing is the most suitable approach for a conversational railway booking system because it can process incomplete and ambiguous passenger requests while reducing unnecessary parsing work.

Combining Earley Parsing with semantic constraints can correctly identify **Chennai → Delhi, two tickets, parents, and AC sleeper class.**

Q3. AI-Based Legal Contract Analysis System

a) Syntactic Structure and Ambiguity

Sentence:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

A simplified structure is:

$S \rightarrow NP VP$

$NP \rightarrow Det N RelClause$

$RelClause \rightarrow who VP$

$VP \rightarrow V NP PP$

Main structure:

[S

[NP The supplier

[Relative Clause who delivered
[NP the equipment]
[PP to the company]]]
[VP must replace
[NP the defective components]
[PP within thirty days]]]

The main sources of ambiguity are:

1. The relative clause **“who delivered the equipment to the company.”**
2. The PP **“to the company.”**
3. The system may incorrectly associate “the company” with the action “delivered.”
4. The long sentence contains multiple noun phrases and prepositional phrases.

The correct interpretation is that **the supplier delivered the equipment to the company.**

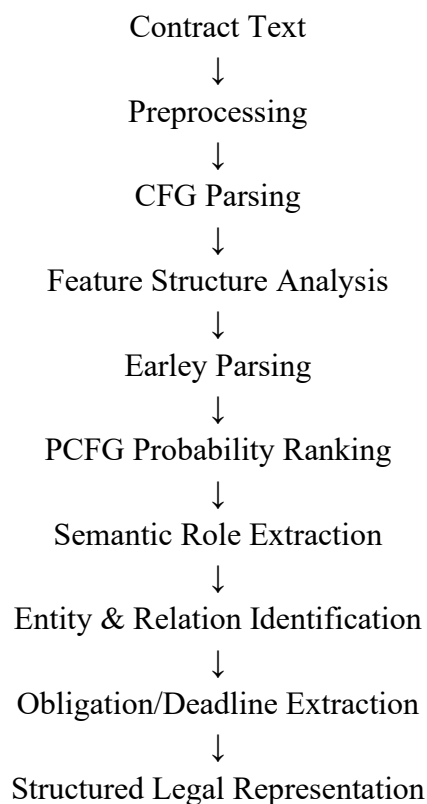
b) Limitations of Basic CFG

Basic CFG is insufficient because:

1. It can generate several structurally valid parse trees.
2. It does not inherently determine the semantic relationship between entities.
3. Relative clauses can create long-distance dependencies.
4. Multiple nested phrases increase parsing complexity.
5. CFG alone does not effectively enforce agreement and semantic constraints.
6. It cannot reliably determine which entity performs an action without additional semantic processing.

Therefore, legal contract analysis requires a combination of syntactic and semantic techniques.

c) Proposed NLP Architecture



CFG: Provides the basic grammatical structure.

Feature Structures: Maintain grammatical features and agreement.

Earley Parsing: Handles complex, nested, and ambiguous structures.

PCFG: Assigns probabilities to competing parse trees and selects the most likely structure.

Semantic Extraction: Identifies parties, actions, obligations, conditions, and deadlines.

d) Step-by-Step Processing Workflow

Step 1: Identify the Supplier

“The supplier” → **Supplier**

Step 2: Identify the Action

“delivered” → **Delivery action**

“must replace” → **Replacement obligation**

Step 3: Identify Equipment

“the equipment” → **Equipment**

Step 4: Identify Company

“the company” → **Company**

Step 5: Identify Obligation

“must replace the defective components” → **Supplier's obligation**

Step 6: Identify Deadline

“within thirty days” → **Deadline = 30 days**

Structured Representation

Supplier → delivered → Equipment

Supplier → delivered Equipment → to Company

Supplier → must replace → Defective Components

Replacement → Deadline → 30 days

Final Extracted Information

Entity/Concept	Extracted Information
Supplier	The supplier
Action	Delivered; must replace
Equipment	The equipment
Company	The company
Obligation	Replace defective components
Deadline	Within thirty days

Justification

The proposed architecture improves contract analysis by combining syntactic parsing with probabilistic and feature-based constraints. Earley Parsing handles nested structures, PCFG selects the most probable interpretation, and semantic extraction identifies the actual responsible party and obligation. Hence, the system reduces incorrect interpretations and improves automated legal contract analysis.

Q4. AI-Based E-Commerce Product Search System

a) Syntactic Ambiguity Using CFG

Query:

“Find laptops with a touchscreen for students under 60,000.”

A simplified CFG is:

$S \rightarrow VP$

$VP \rightarrow V NP$

$NP \rightarrow N PP PP$

$PP \rightarrow P NP$

Possible Interpretation 1:

Find [laptops] [with a touchscreen] [for students] [under ₹60,000].

Meaning:

- Product = Laptop
- Attribute = Touchscreen
- Target users = Students
- Maximum price = ₹60,000

Possible Interpretation 2:

Find [laptops with a touchscreen for students] [under ₹60,000].

Here, “**for students**” may be treated as part of the description of the touchscreen/laptop phrase.

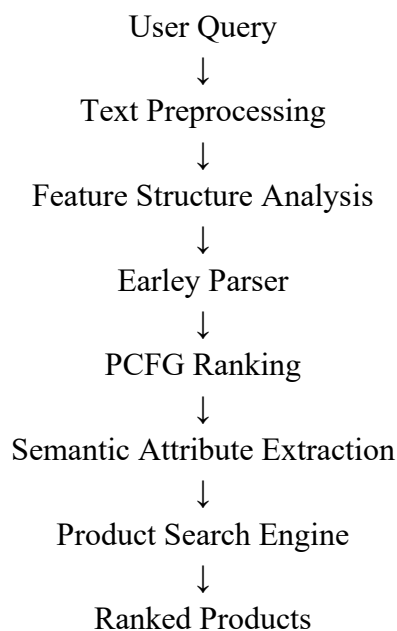
The correct interpretation is that “**for students**” **describes the laptops**, not the touchscreen.

b) Limitations of CFG

Basic CFG has the following limitations:

1. It may create multiple interpretations for ambiguous phrase attachments.
2. It does not handle missing words effectively.
3. Informal customer queries may not follow complete grammar rules.
4. Long product descriptions increase parsing complexity.
5. Coordination and elliptical expressions can cause incorrect structures.
6. CFG does not naturally rank different interpretations.

c) Improved Parsing Architecture



Feature Structures identify:

- Product type = Laptop
- Screen feature = Touchscreen
- User requirement = Students
- Price constraint = $\leq$ ₹60,000

Earley Parsing handles incomplete and ambiguous queries.

PCFG ranks alternative parse structures and selects the most probable one.

d) Justification

The proposed system improves:

Search Precision: Correctly separates product attributes, user requirements, and price constraints.

Response Time: Earley Parsing avoids unnecessary repeated parsing and supports partial input.

Customer Experience: Natural conversational queries can be understood without requiring predefined

filters.

Scalability: The architecture can process large numbers of customer queries and product descriptions. Therefore, combining PCFG, Feature Structures, and Earley Parsing provides more accurate product search results.

Q5. Intelligent Airline Voice Assistant

a) Possible Structures and Ambiguities

Command:

“Change my flight to Mumbai tomorrow with two passengers.”

Possible interpretation 1:

Change [my flight] [to Mumbai] [tomorrow] [with two passengers].

Meaning:

- Destination = Mumbai
- Date = Tomorrow
- Passengers = 2

Possible interpretation 2:

Change [my flight to Mumbai] [tomorrow with two passengers].

Here, the phrase “**with two passengers**” may be incorrectly associated with the flight/destination.

The correct interpretation is:

Flight → **Mumbai**

Date → **Tomorrow**

Passengers → **2**

Another command is:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

This contains a **self-correction**, where Bangalore replaces Mumbai.

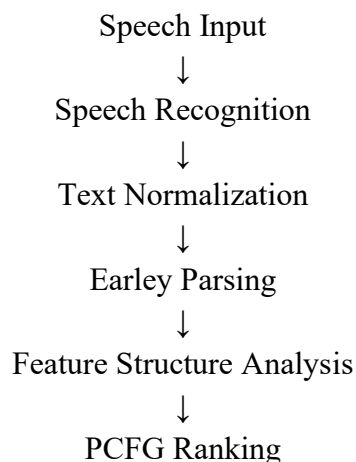
b) Limitations of Top-Down Parsing

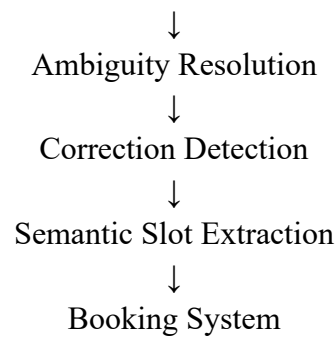
Top-down parsing faces several problems:

1. It may perform excessive backtracking when multiple structures are possible.
2. It may incorrectly attach “with two passengers.”
3. Incomplete speech can leave unfinished parse structures.
4. Speech recognition errors may produce incorrect words.
5. User corrections require the previous interpretation to be updated.
6. Repeated parsing can increase response delay.

Therefore, conventional top-down parsing is not ideal for real-time voice assistants.

c) Proposed Architecture





Earley Parsing: Handles partial and incomplete spoken commands.

PCFG: Ranks competing interpretations.

Feature Structures: Enforce constraints such as number and agreement.

Correction Detection: Identifies that “Bangalore” replaces the previously mentioned “Mumbai.”

d) Processing Workflow

Step 1: Speech Input

“Change my flight to Mumbai tomorrow with two passengers.”

Step 2: Speech Recognition

Convert speech into text.

Step 3: Parsing

Earley Parser generates possible structures.

Step 4: Ambiguity Resolution

PCFG and semantic constraints select:

- Destination = Mumbai
- Date = Tomorrow
- Passengers = 2

Step 5: Correction Processing

Input:

“Actually, make that Bangalore.”

The system identifies a correction:

Previous destination = Mumbai

New destination = Bangalore

Therefore:

Destination = Bangalore

Step 6: Structured Booking Request

Field	Value
Action	Change flight
Destination	Bangalore
Date	Tomorrow
Passengers	2

Final Justification

The proposed architecture is suitable for real-time airline applications because Earley Parsing supports partial and ambiguous input, PCFG helps select the most probable interpretation, and Feature Structures enforce grammatical constraints. Correction detection allows the system to update previously extracted information. This improves **accuracy, response speed, robustness, and user experience** in voice-based flight booking.